9.0 Introduction

The various parts of the vehicle are mounted on the chassis frame. The chassis frame must strong enough to withstand total weight of the vehicle. The chassis frame carries engine, clutch, gear box, Propeller shaft, rear axle, suspension, steering gear box and brake. Frame is fixed with two longitudinal bars and four or five cross bars by rivet or bolts.

9.1 Chassis frame

- The skeleton of the vehicle is chassis frame.
- Chassis frames are made up of alloy steel and it is of high strength to make the rigid structure.
- Based on the construction, application, load capacity of the vehicle, the chassis frame is made up of different size and shape.
- Chassis frame forms the backbone of the vehicle.

9.1.1 Functions of the chassis frame

1. To withstand the weight of the vehicle, passenger and goods.
2. To form the base to mount the engine, suspension, steering and power transmission system.
3. To build the vehicle body on the chassis frame.
4. To withstand the reaction due to driving and braking force.
5. To absorb the ground reaction forces due to road undulations.
6. To withstand the static load and dynamic load of a vehicle.

9.1.2 Types of chassis frame

Chassis frame are broadly classified into three types, namely:

1. Conventional Frame
2. Semi Integral Frame
3. Integral or Unit Frame

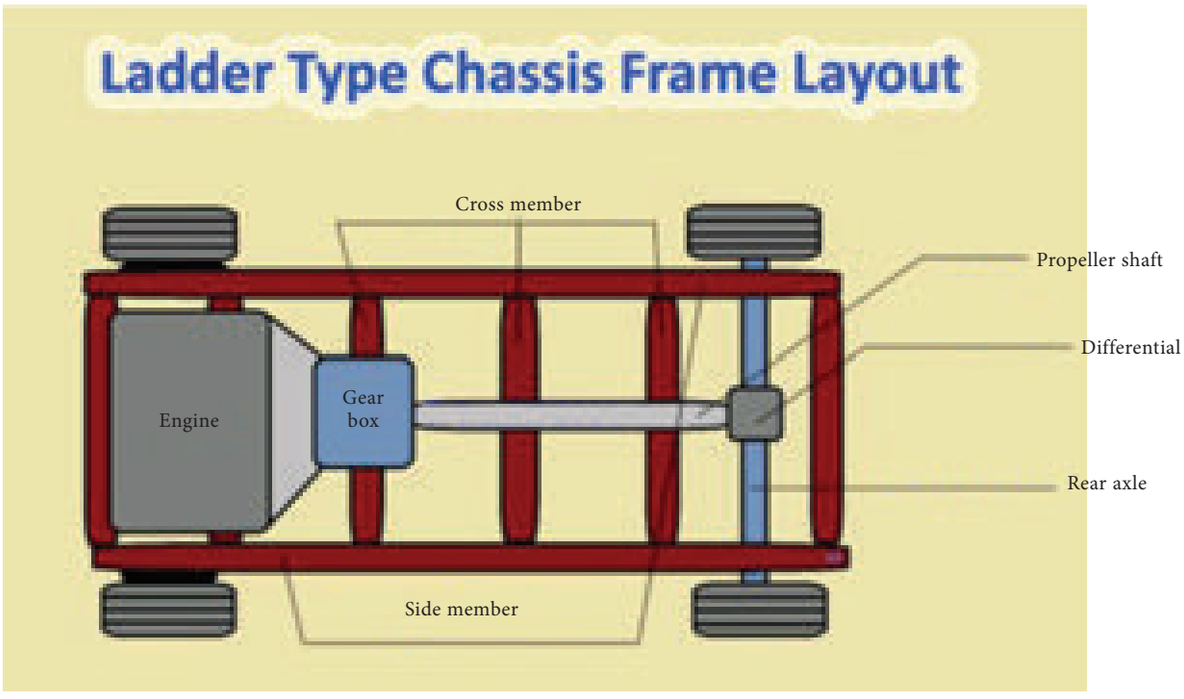


Figure 9.1 Ladder type chassis frame

BAE_ENG_Chapter_09.indd 168 12/20/2019 1:05:54 PM